# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-COVERAGE-EN · v2.1 · 2026-11-01 |
| Package | star-coverage |
| Version | 0.1.0 |
| Capability | star.orbit.coverage-geometry |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_coverage |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_coverage-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
| --- | --- |
| star_coverage-0.1.0-py3-none-any.whl | 13bc7a824322eed49271530096f158ee79db961e2ac4b754d410b82ed9ff4466 |
| star_coverage.py | b7bf0970d36692f5579c8462cc54e73ae389b2df0844b1fcc69157a30f3dcfad |
| star_coverage-0.1.0-source.zip | 555c1a7e8fb0cbb249682145378d03b01583d9f71e98cdb1c54754b009d72859 |

# 5. What it does, and its limits

Earth-central angle, nadir angle, slant range, covered fraction, footprint area and elevation for a satellite at a given altitude and minimum elevation; agrees with pymap3d and PROJ on a sphere to 1e-13 and keeps precision from 1 mm of altitude to 1e12 km.

A sphere: on the real Earth the horizon distance changes by up to about 0.3 % with latitude. No refraction, terrain or antenna pattern; no constellation design (number of satellites, overlap, revisit).

- `central_angle_deg`, `nadir_angle_deg`, `slant_range`, `coverage_fraction`, `footprint_area` for an altitude and a minimum elevation; `elevation_deg(altitude, central_angle)` is the inverse relation (negative below the horizon). Default radius 6378.137 km; `radius=` for another body or unit. The three angles add up to 90 degrees.
- Reproduces the textbook geostationary figures (81.3 deg, 8.7 deg, 41 679 km, 42.4 % of the surface, 76.3 deg at 5 deg of elevation) and the 2294 km horizon of a 400 km orbit.
- Measured against pymap3d and PROJ on a sphere (400 cases, 200 km to 400 000 km): elevation within 8e-14 deg, slant range within 4e-15 relative, central angle within 3e-14 deg. The formulas are written to keep precision at both ends: 1 mm above the ground the slant range is right to 1e-14 (the textbook form loses seven digits there), and at 1e12 km the central angle stays below 90 degrees.
- Every function returns a finite float or raises `ValueError`: non-numeric, boolean, NaN or infinite input, altitude <= 0, radius <= 0, minimum elevation outside [0, 90], central angle outside [0, 180], an altitude so small or so large against the radius that the ratio is not representable.

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_coverage_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-coverage}: What a satellite sees, and from where it is seen},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_coverage/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts